

ELI — Diagrams

ELI v2.3.72

September 2026

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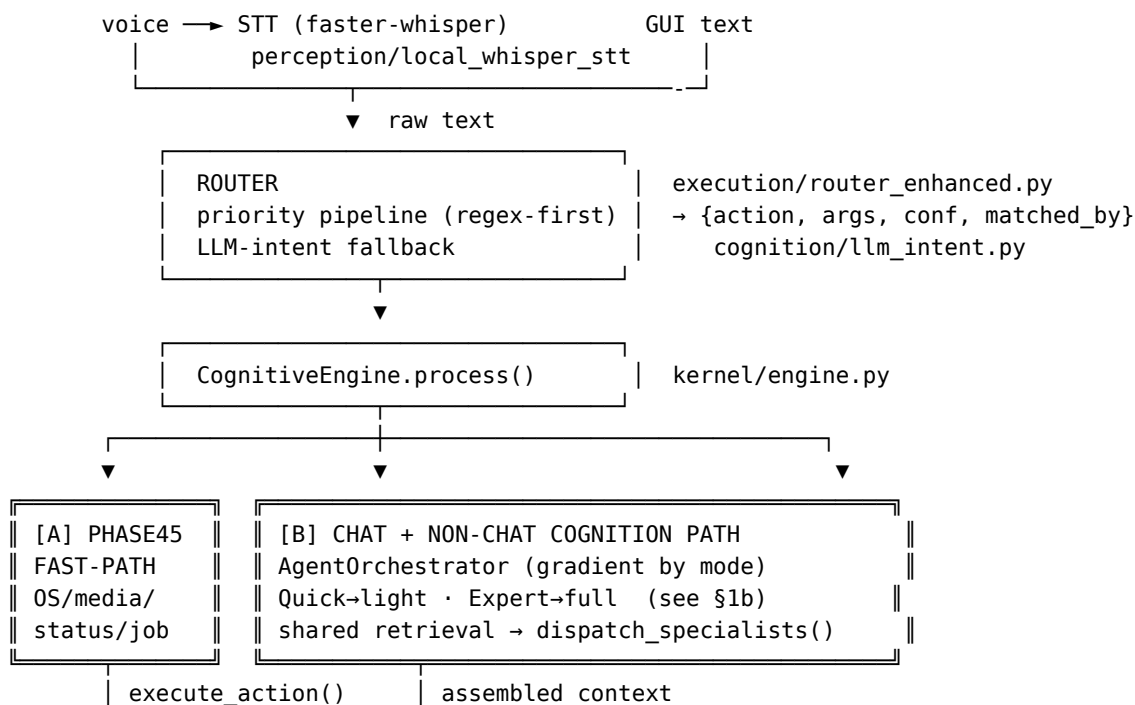
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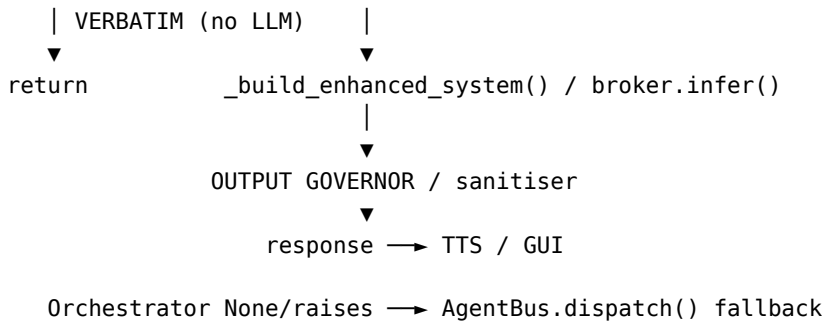
Blueprint — ELI MKXI ASCII Diagrams

Updated for v2.3.72. CHAT uses a gradient orchestrator for all modes; retrieval is unified in eli/memory/retrieval.py; Stage 12 = learning/state commit.

Visual companion to architecture.md. Three views: the full request pipeline, the memory subsystem, and the gating stack. All grounded in the real modules (file paths are clickable).

1. Full pipeline (request lifecycle)

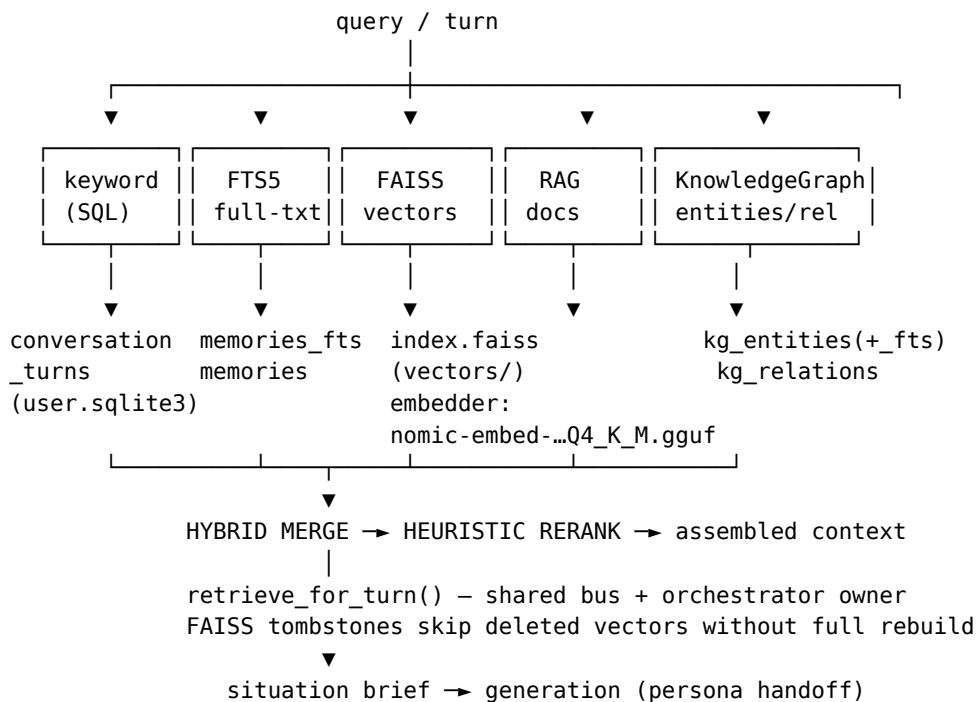




1b. Orchestrator — canonical S01-S12 (pipeline_trace.py)

S01 PERCEIVE → S02 INPUT_GUARDS → S03 ROUTER → S04 GROUNDING_GATE
 → S05 PLANNER (mode-aware budgets)
 → S06 AGENT_BUS: retrieve_for_turn() + dispatch_specialists()
 → S07 CONTEXT_ASSEMBLY (merge + heuristic rerank)
 → S08 INFERENCE_BROKER → S09 REASONING_SYNTHESIS (mode algo)
 → S10 OUTPUT_GOVERNOR → S11 RESPONSE_DELIVERY
 → S12 LEARNING_STATE_UPDATE (learning_coordinator.finalize_turn)

2. Memory subsystem eli/memory + cognition/working_memory.py



STORES (artifacts/)	
db/user.sqlite3	memories(+_fts), conversation_turns, conversations, kg_entities(+_fts), kg_relations, recall_log, runtime_events, news_articles(+_fts), news_reflections, habits/habit_events/habit_rules, observations, learning_replay, working_memory_pins, user_patterns, session_summaries, corrections, failures
db/agent.sqlite3	agent_dispatches, agent_metrics, improvements,

failures, code_patches, error_tracking, observations
vectors/index.faiss semantic index
runtime/users/<uuid>/user_profile... per-user profile

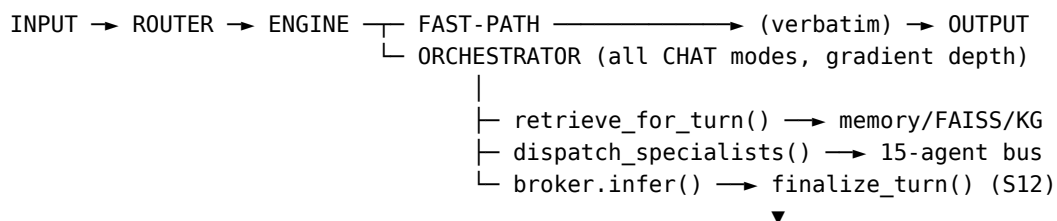
WRITE PATH: turn → PERSISTENCE GATE (drop junk/report-dumps) → store

3. Gating stack (in path order — every guard the request passes)

1	NETGUARD (process-wide, always-on)	core/netguard.py
	any outbound socket → offline? → OfflineError (FAIL-CLOSED)	
	allow_network(): scoped opt-in window (model download, web tier)	
2	PHASE45 FAST-PATH GATE	kernel/engine.py
	deterministic action (volume/media/date/job/status)?	
	→ execute_action() VERBATIM, no LLM (cannot confabulate)	
3	PERSISTENCE GATE	runtime/persistence_gate.py
	about to write to memory? junk / internal report-dump?	
	→ DROP (so it can't be stored and replayed as fact later)	
4	GROUNDING GATE	runtime/deterministic_gate.py
	answer requires evidence? unbacked or raw packet?	
	→ BLOCK raw evidence/telemetry leak; require grounded synthesis	
5	GROUNDING ESCALATION	runtime/grounding_escalation.py
	checkable fact AND grounding < threshold ?	
	external fact → WEB tier	first tier that grounds → answer
	self/project → LOCAL tier	
	exhausted / offline → HEDGE ("I won't guess")	
	(trigger = GROUNDING, not the response score – which lies when wrong)	
6	LEARNING / STATE COMMIT (S12)	learning_coordinator.py
	finalize_turn(): store assistant turn · publish meta · learn from result	
7	OUTPUT GOVERNOR / SANITISER	cognition/output_governor.py
	final text → strip leaks, persona hygiene, no raw JSON → user	

Legend: FAIL-CLOSED = denies by default; VERBATIM = returned without an LLM pass; HEDGE = honest "can't verify" instead of a guess.

4. One-screen overview (how it all connects)



GATES: netguard · persistence · grounding · governor



OUTPUT



background: proactive · self-improve · habits ·
learning(LoRA) · world · scheduler

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